

FEDERAL POLYTECHNIC MUBI

SCHOOL OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

NATIONAL DIPLOMA PROGRAMME

COURSE:

MACHINE TOOLS TECHNOLOGY AND PRACTICE

COURSE CODE:

MEC 123

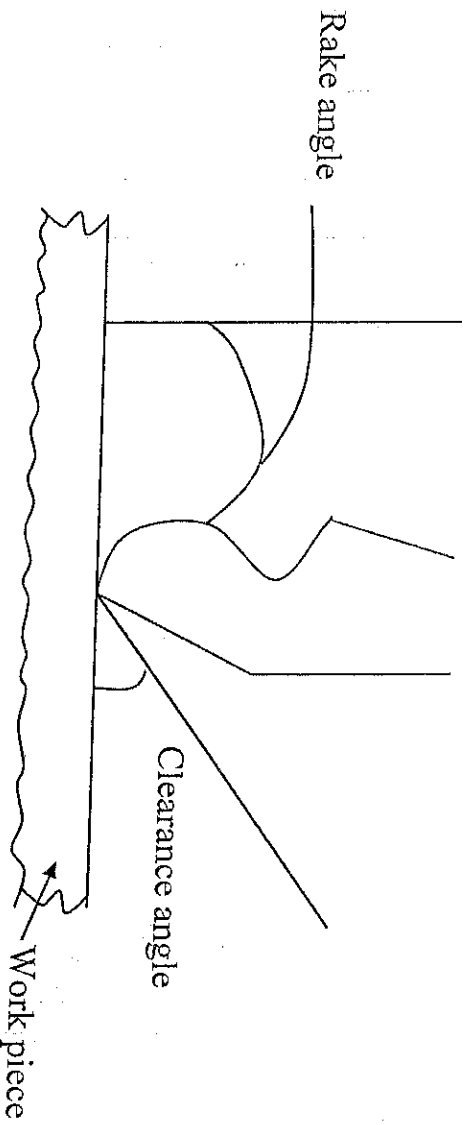
THEORY OF METAL CUTTING

Introduction

When we discuss the topic machine tool we are simply referring to those machine in use in the process of production using parts of machine and equipment in the production workshop, factories and industries. Example of such machines are as follows: Lathes, shapers, mining machines. Grinding machines engraving machines and planners etc.

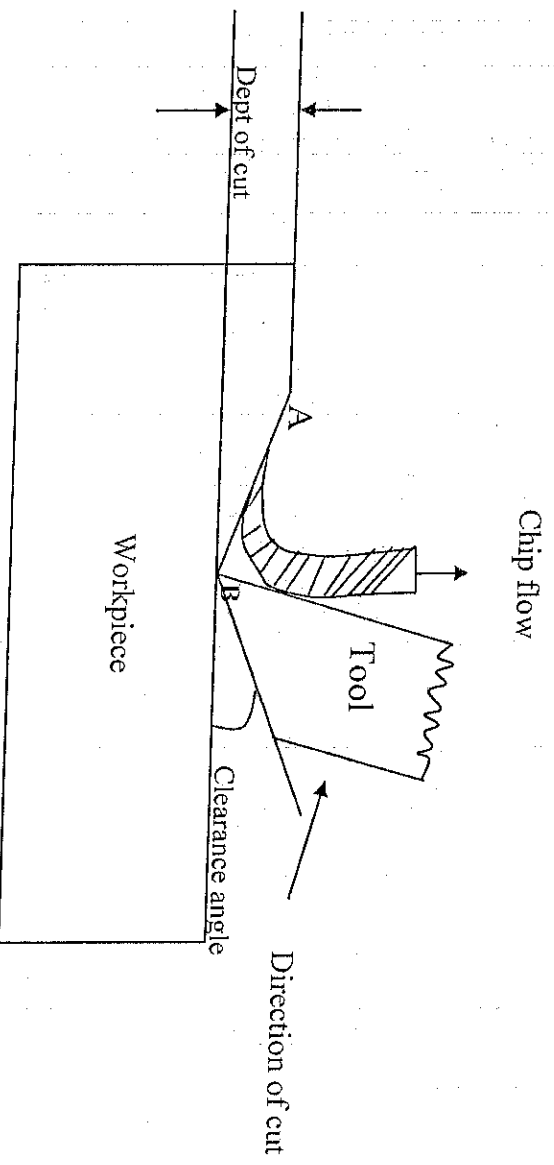
In the process of employing the above-mentioned machines to achieve our main objectives, it is mandatory that certain tools must be put in use such tools are referred to as cutting tools. These tools are categorized into two which are single and multi-point tools.

Though the main aim of all machine tools is to ensure that essential geometrical movements of work and tools are achieved, yet the actual removal of metal chips is carried out by the cutting tools in use. It is obvious that cutting tools should either be single-point tool have wedge like action while the multi-point ones are made up of number of wedge-like points.



The usual conception of cutting suggest clearing the substances apart with a thin knife or wedge. When metal is cut the action is rather different and although the tool will always be wedge shape in the cutting area and the cutting edge should always be +sharp, the wedge angle will be far too great for it to be for it to be considered knife shaped. Consequently, a shaping action take place when the work moves against the tool.

The diagram below shows a tool being moved against a fixed work piece. When the cut is in progress the chip presses heavily on the top face of the tool and the continuously shearing takes place across the shear plain AB although the diagram shows it working in the horizontal plain in the work piece and the same action takes place with the work piece revolving and the tool stationary.



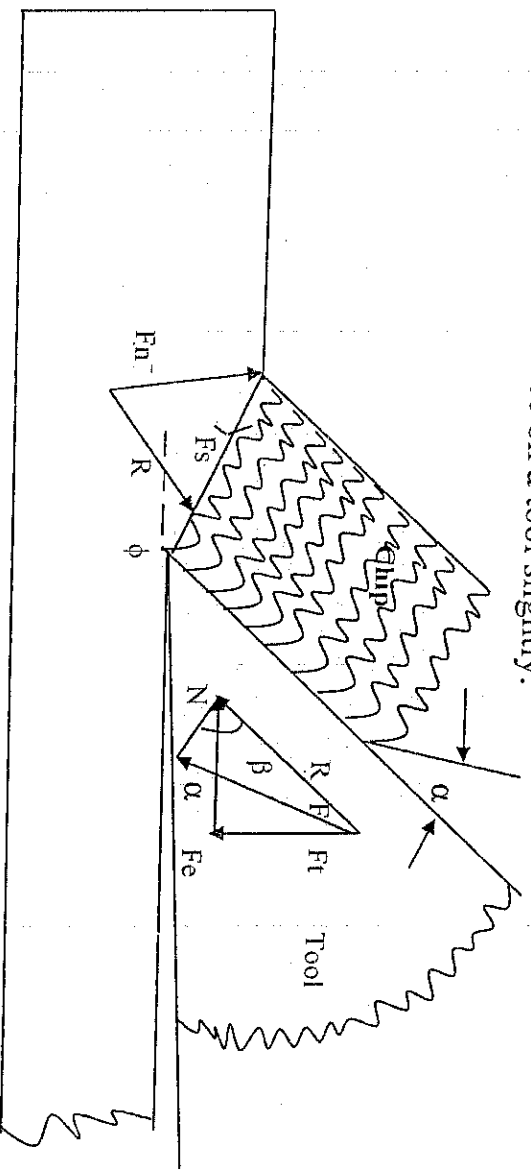
FORCE ACTING ON THE TOOL POINT

During every machine operation in the production process, most of the force experienced result into frictional force but during the cutting process on machine tools, the bearing is more of shearing and tearing forces. These type of forces

however determine the result or the finish on every product from machining operations.

These cutting force is the most essential. The force on a cutting tool a given material depend on a number of consideration.

- (i) Tool force are not changed by a change in cutting speed
- (ii) The great the speed of the tool, the larger the forces.
- (iii) The greater the depth of cut, the large the forces.
- (iv) Cutting forces increases with chip sizes.
- (v) Feed force is decreased if nose radius made larger.
- (vi) Using a coolant reduces the force on a tool slightly.



Forces acting at a cutting tool point

Force acting on a cutting tool in two-dimensional cutting.

NB:

- (i) Resultant forces, R must be collinear to balance the force.
- (ii) The simplest cutting operation is one in which is a straight-edged tools moves with a constant velocity in the direction perpendicular to the cutting

edge of the tool, this is known as the Orthogonal cutting process or dimensional.

TOOLS SURFACE AND ELEMENTS

The design component of the cutting tool are defined as follow:

- ✓ Rake face is the surface over which the chip, formed in a cutting-sliders.
- ✓ Flank face is the surface(s) over which the surface, produced on the work piece passes.
- ✓ Cutting edge is a theoretical line of intersection of the rake and the flank surface.
- ✓ Cutting wedged is the tool body enclosed between the rake and flank faces
- ✓ Shrank is the part of the tool by which it is held.

TOOL ANGLES

There are three important angles in the construction of a cutting tool rake angle, clearance angle and plan approach angle.

Rake Angle

Rake angle is the angle between the top face of the tool and the normal to the work surface at the cutting edge. In general, the larger the rake angle, the smaller the cutting force on the tool, since for a given depth of cut the shear plane AB shows diagram 1. Decrease at rake angle increases. A larger rake angle will improve cutting action, but would lead to early tool failure, since the tool wedge angle is relatively weak. A compromise must therefore be made between adequate strength and good cutting action.

Table 1: Typical Angle For Top Rake Angle.

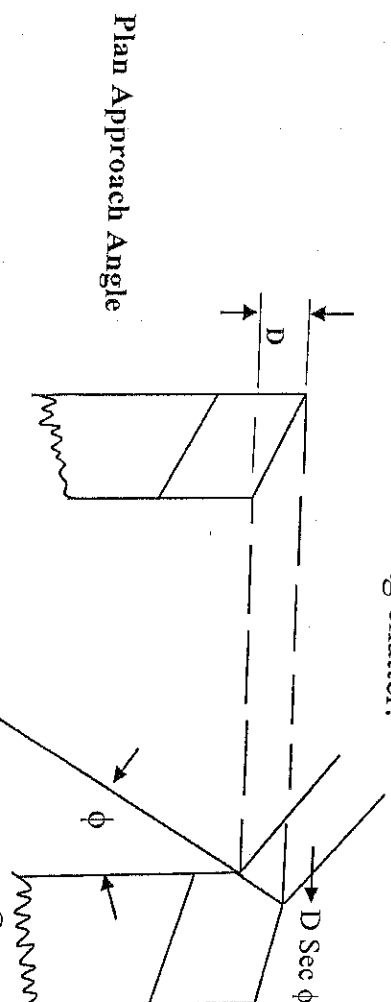
Metal being cut	Cast iron	Hard steel brass	Medium carbon steel	Mild steel	Aluminum
Top rake angle	0°	8°	14°	20°	40°

Clearance angle

Clearance angle is the angle between the flank or front face of the tool and the tangent to the work surface originating at the cutting edge. All cutting tools must have clearance to allow cutting to take place. Clearance should be kept at a minimum, as excessive clearance angle will not improve cutting efficiency and will merely weaken the tool. Typical value for front clearance angle is 6° in external turning.

Plan Profile of Tool

The plan shape of the tool is often dictated by the shape of the work, but it also has an effect on the tool life and the cutting process. The diagram below shows the two tools, one where a square edge is designed and the other where the step in the work end with a chamfer or angle. The diagram shows that, for the same depth of cut, the angled tool has a much greater length of cutting edge in contact with the work and thus the load per unit length of the edge is reduced. The angle at which the edge approaches the work should in theory be as large as possible, but if too large, chatter may occur. This angle known as the plan approach angle, should therefore be as large as possible without causing chatter.



The angle ground on a lathe tool to give the necessary shape for efficient cutting will depend on:

1. The hardness of the material being cut.
2. The rigidity of the lathe.
3. The amount of metal to be removed.
4. The quality of surface finish required.

A Large Tool Angle will:

- (i) Give maximum strength.
- (ii) Help heat dissipation.
- (iii) Increase the power needed to force the tool into the work.

A Small Tool Angle Lathe will:

- (i) Weaken the cutting edge.
- (ii) Gives a good finish.
- (iii) Gives little tearing.
- (iv) Gives easy cutting.

CHIP

In the context of machining, a cutting tool or cutter is any tool that is used to remove materials from the workpiece by means of shear deformation. That material removed from the workpiece is called Chip.

TYPES OF CHIPS

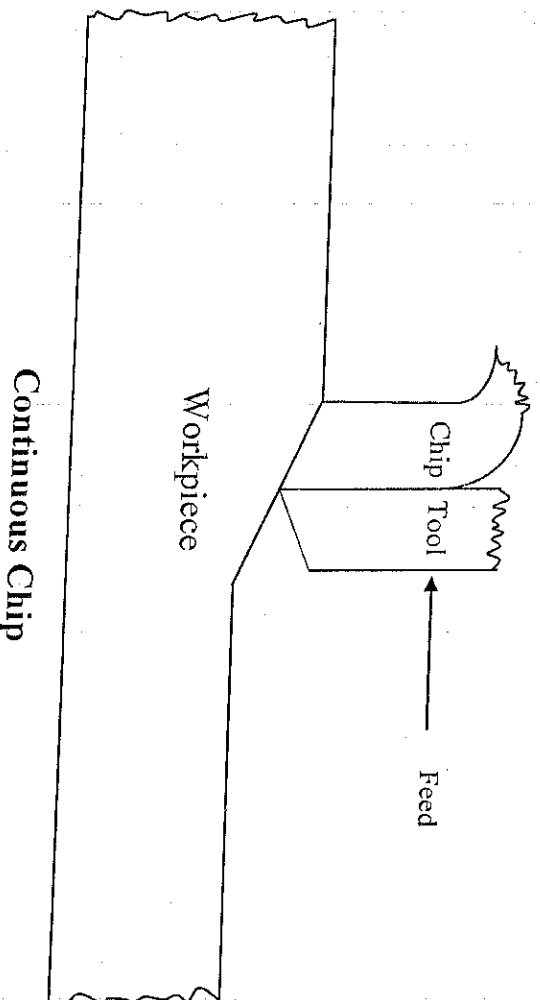
The three common types of Chip are:

- (a) **Continuous Chips:** Continuous Chips are formed by the continuous plastic deformation of metal without fracture in front of the cutting edge of the tool and is formed by the smooth flow of the chip up the tool face. Mild steel and Copper are considered to be most desirable materials for obtaining continuous chips. The chips

obtained have some thickness throughout. This type of chip is the most desirable. Since it is stable cutting, resulting in generally good surface finish. On the other hand these chips are difficult to handle and dispose off.

Continuous Chips tend to be formed when the following condition exist:

- (i) Ductile material.
- (ii) High cutting speed.
- (iii) Small chips thickness.
- (iv) Large rake angle.
- (v) Minimum friction of chip on tool face by
 - Polished tool face.
 - Use of efficient cutting lubricant.
 - Use of tool material with low coefficient of friction.

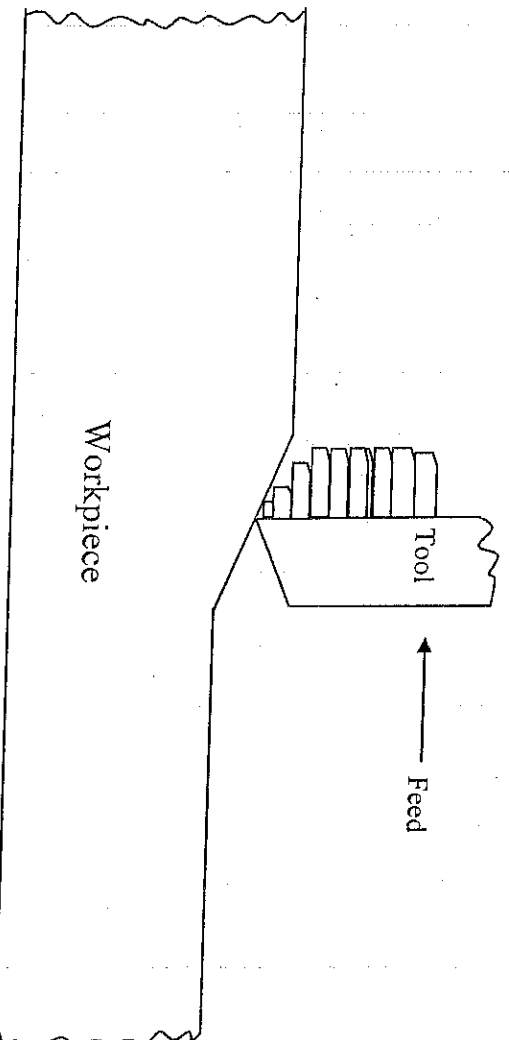


(b) **Discontinuous or Segmental chip:** Discontinuous chip is formed by a series of rupture (breaking) occurring approximately perpendicular to the tool place face' each chip element passing off along the tool face, the chip element' in the form of small segment may adhere loosely to each other and becomes slightly longer.

Since the chips breakup into small segments the friction between the tool and the chips reduces resulting is better surface finish. These chips are convenient to collect handle and dispose off. Discontinuous chips tend to be formed when one or more of the following conditions exists:

- (i) Brittle material, such as cast iron and bronze.
- (ii) Large chip thickness.
- (iii) Low cutting speed.
- (iv) Small rack angle.

Discontinuous chips are also produced when cutting more ductile material with the use of a cutting fluid.



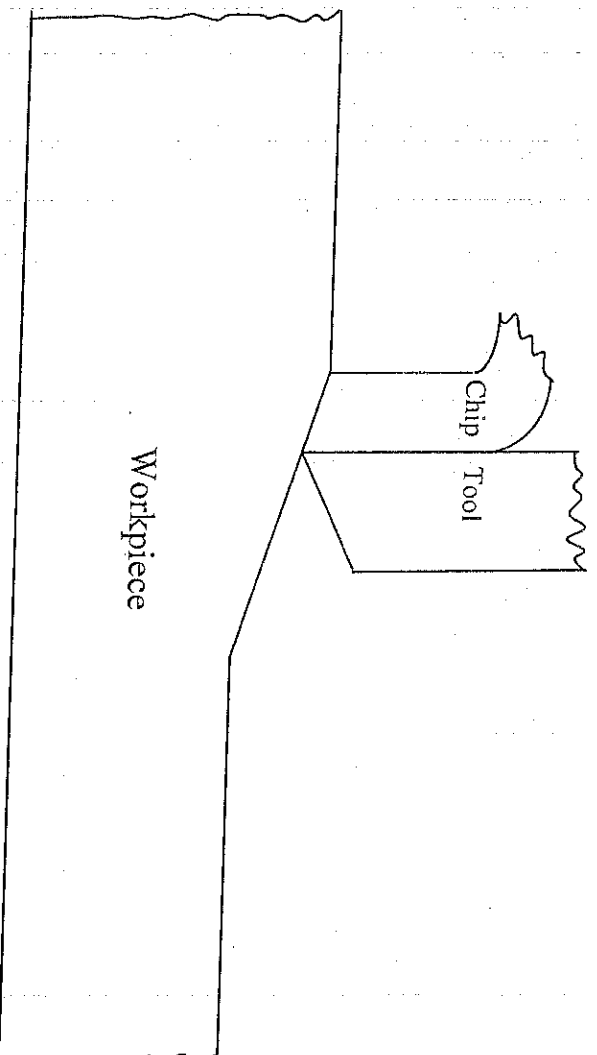
Discontinuous Chip

(c) **Continuous Chip with Built Up Edge:** This type of chip is very similar to the continuous chip. With the difference that it has a built-up edge adjacent to face and also it is not so smooth. It is obtained by machining on ductile material, in this condition of high local temperature and extreme pressure in the cutting and high friction in the tool chip interference, may cause the work material to adhere or weld

to the cutting edge of the tool. Successive layers of work material are then added to the built-up edge. When this edge changes its size during the cutting operating. It first increases then decreases, then again increases etc.

The following conditions favours the formation of continuous chips with built up edge (BUE):

- (i) Work material-ductile.
- (ii) Cutting velocity-low ($\mu 0.5\text{m/s}$).
- (iii) Small or negative rake angles.
- (iv) Cutting fluid-inadequate or absent.
- (v) Feed-medium or large.



Continuous Chip with Built-up Edge

Factors Affecting The Types of Chips

There are several factors which can affect the chips formation:

- (i) The type or nature of metal being worked upon.
- (ii) The amount of clearance angle on the tool in use.
- (iii) The depth of cut that is whether it is light or heavy cut.

- (iv) The nature of lubricant or coolant in use which of course depends on the metal constituent.
- (v) The surface speed of the machine in use i.e. RPM.
- (vi) The feed taken which is the amount of cut in a given revolution.

CUTTING FLUIDS

HEAT GENERATED DURING CUTTING

In the process of cutting metals heat is generated by plastic deformation of metal which occurs immediately ahead of the cutting tool, accounts for approximately two-thirds ($\frac{2}{3}$) to three-quarters ($\frac{3}{4}$) of the heat generated.

The friction created by the chip sliding along the cutting tool face, the cutting temperature varies with each type of metal and increases with the cutting speed employed, and the rate of removal. The greatest heat is generated when ductile material with high tensile strength, such as steel is cut while the lowest heat is generated when a soft material of low tensile strength such as aluminum is cut.

The heat generated at the chip tool interface must find its way into one of three places, the workpiece, the cutting tools or the chips. If the workpiece receives too much heat (thermal) damage will occur at its surface and also if cutting tool receives the cutting edge will break down rapidly, reducing tool life, the ideal cutting tool is the one that transfer the heat quickly from the cutting zone to some of cooling system and most of the heat is taken off in the chips.

TYPES OF CUTTING FLUID

Cutting fluids falls into three (3) categories:

- (i) Cutting oil.
- (ii) Emulsifying (soluble oil).
- (iii) Chemical (synthesis cutting fluid).

(1) **Cutting Oil:** These are classified into active or inactive. These terms relate to the oil's chemical activity or ability to react with the metal surface at elevated temperature, to protect it and improve cutting action.

(2) **Emulsifiable (soluble oil):** These are mineral oil's containing a soap-like material that makes them soluble in water and causes them to adhere to the workpiece during machining.

(3) **Chemical Cutting (Synthetic):** Fluid have been widely accepted and used for many types of metal-cutting operation. They are stable, performed emulsions that contain very little oil and mixed easily with water some of the chemical cutting fluid contain extreme pressure (EP) lubricant which can reduce heat of friction between the chip and tool face and heat caused by plastic deformation of metal.

FUNCTIONS OF CUTTING FLUID

- (i) Cooling
- (ii) Lubrication
- (iii) Cutting tool life
- (iv) Rust control
- (v) Rancidity control
- (vi) Improve good surface finish
- (vii) Energy requirements

- **Cooling:** It is to reduce the cutting-tool temperature, over a small reduction in cutting-tool temperature will greatly extend the life of a cutting tool and also help to wash the chips away from machining area and increase cutting speed.
- **Lubrication:** This is to reduce the friction between the chip and tool face, the effective life of cutting tool can be greatly lengthened, if the friction and the resultant heat generated are reduced.
- **Cutting Life Tool:** The use of an effective cutting fluid affects the action of a cutting tool by lowering the heat created, reducing the friction, and reducing the power required for machining.

- **Rust Control:** Cutting fluids used on machine tools should prevent rust from forming. Otherwise, machine parts and workpiece with damaged.
- **Rancidity Control:** The rancidity is caused by bacteria and other organism growing and multiplying almost everywhere and eventually causing bad odour to form. Most cutting fluids contain bactericides that control growth of bacteria and make the fluid more resistant to rancidity.
- **Improve Good Surface:** The presence of cutting fluids helps to improve good surface-finish on manufactured parts.
- **Energy Requirements:** It helps to reduce energy requirement during the cutting/ manufacture operation.
- It helps to carry away swerves and chips from the tools.

CHARACTERISTICS OF CUTTING FLUID

For a cutting fluid to friction effectively it must possess the following desirable characteristics.

1. **Good Cooling Capacity:** To reduce the cutting temperature, increase tool life and production and improve dimensional accuracy.
2. **Good Lubricating Qualities:** To prevent metal from adhering to the cutting edge and forming a built-up edge, resulting in a poor surface finish and shorter tool life.
3. **Rust Resistance:** To prevent stain, rust or corrosion to the workpiece or machine.
4. **Stability (Long life):** To holdup in storage and in use.
5. **Resistance to rancidity:** Not to become rancid easily.
6. **Non-Toxic:** Not to cause skin irritation to the operator
7. **Transparent:** To allow operator to see the work clearly during machining.
8. **Relatively Low Viscosity:** To prevent the chips and dirt to settle quickly.

9. Non-Flammable: To avoid burning easily and should preferably be non-combustible.

RECOMMENDED CUTTING FLUID FOR VARIOUS MATERIALS

Materials	Drilling	Reaming	Threading	Turning	Milling
Aluminum	Soluble oil, kerosene and lard oil	Soluble oil, kerosene and mineral oil	Soluble oil, kerosene and lard oil	Soluble oil	Soluble oil, mineral oil
Brass	Dry soluble oil, kerosene and lard oil	Dry soluble oil	Soluble oil, and lard oil	Soluble oil	Dry soluble oil
Bronze	Dry soluble oil and lard oil	Dry soluble oil and lard oil	Soluble oil and lard oil	Soluble oil	Dry soluble oil, lard oil
Cast Iron	Dry soluble oil, lard oil	Dry soluble oil, mineral and lard oil	Dry surface oil mineral lard oil	Soluble oil	Dry soluble oil
Copper	Dry soluble oil, mineral oil and kerosene	Soluble oil, lard oil	Soluble oil and lard oil	Soluble oil	Dry soluble oil
Steel Alloy	Soluble oil, Sulfurized oil, mineral oil	Soluble oil, Sulfurized oil, mineral oil and lard oil	Sulfurized oil, and land oil	Soluble oil	Soluble oil, mineral land oil
Malleable Iron	Dry soda water	Dry soda water	Lard oil and soda water	Soluble oil	Dry soda water

CUTTING TOOLS

The tool used in a lathe is known as a single point cutting tool. It has one cutting edge or point whereas a drill has two cutting edges and a file has numerous point or teeth.

The lathe tool shears the metal rather than cuts as will be seen later and it can only do so if there is relative motion between the tool and the workpiece. Ofcourse the amount of movement is of paramount importance too much at once could for instance result in breakage of the tool.

Linear cutting tools include tool bits (single-point cutting tools) and broaches while rotary cutting tools include drill bits countersink and counterbores, taps and dies, milling cutters, reamers and cold saw blades other cutting tools such as bandsaw blades, hacksaw blades and fly cutters, combine aspects of linear and rotary motion.

PROPERTIES OF CUTTING TOOL MATERIALS

- (i) They should be hard.
- (ii) They should be wear resistance.
- (iii) They should be capable of maintaining a red hardness during machining operation.

NB: Red hardness is the ability of a cutting tool material to maintain sharp cutting edge even when it turns to reduce to the high heat produced at the work tool interface during the cutting operation.

- (iv) They should be able to withstand shock during the cutting operation.
- (v) They should be shaped so that the edge can penetrate the work.
- (vi) They should possess high compression of a materials will take before it repture or breaks.
- (vii) They should have excellent thermal conductivity.

TYPES OF CUTTING TOOLS

1. Rough left hand tool.
2. Rough right hand tool.
3. Round nose finishing tool.
4. Parting-off tool.
5. Form tool.
6. Thread cutting tool.
7. Boring tool.

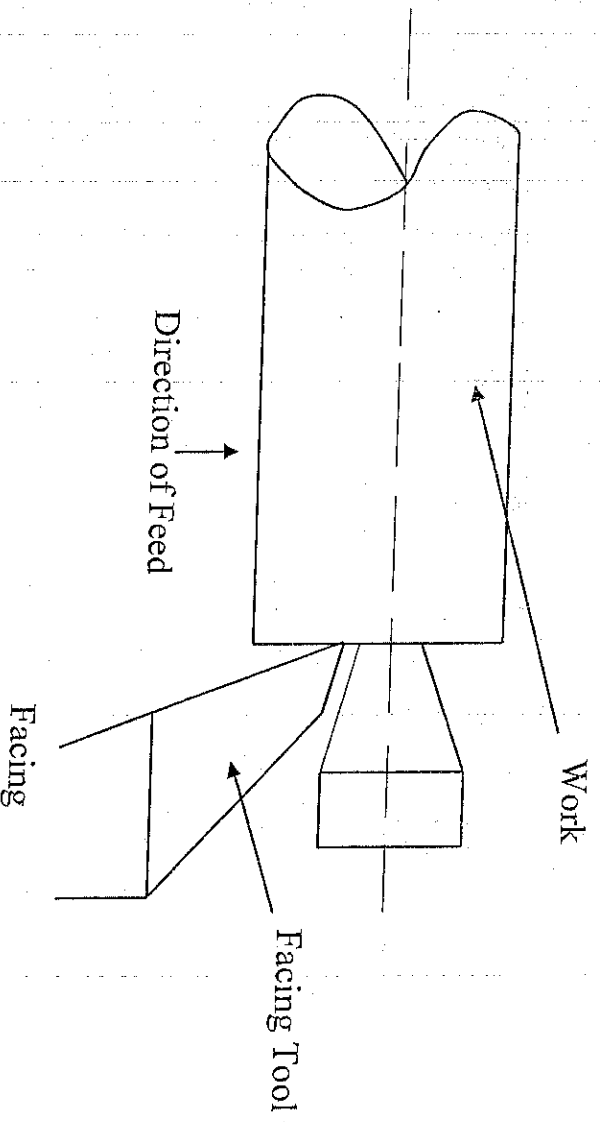
MACHINING OPERATIONS ON THE CENTRE LATHE

Various types of operations performed in Lathe machine

Various operations are performed in Lathe machine other than plain turning. These are:

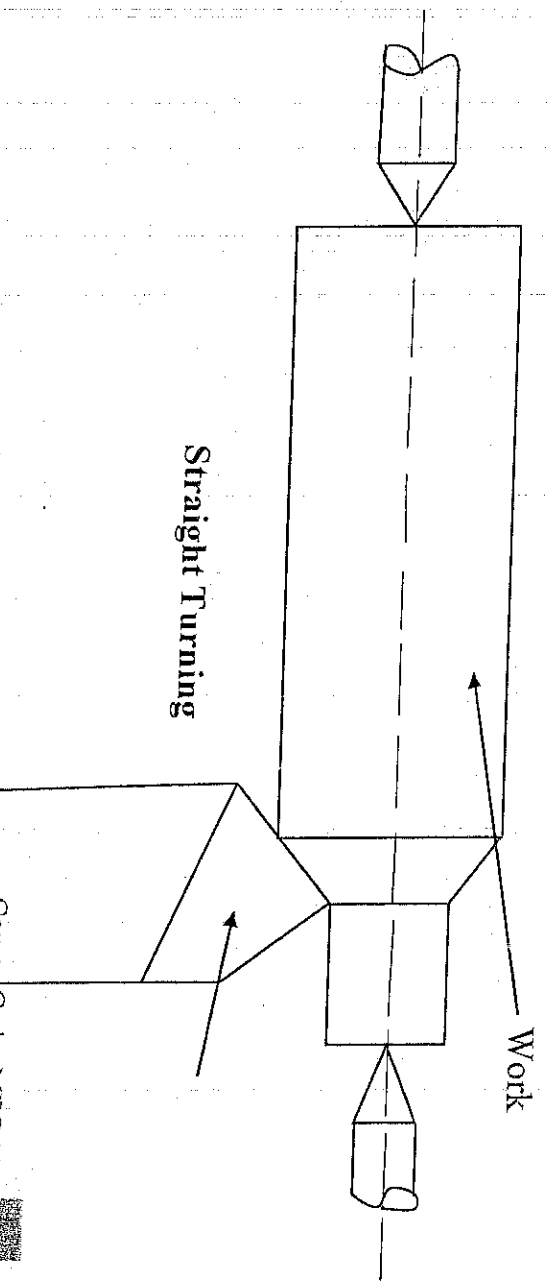
1. Facing
2. Turning: (a) Straight turning (b) Step turning.
3. Chamfering.
4. Grooving.
5. Forming.
6. Knurling.
7. Undercutting.
8. Eccentric turning.
9. Taper turning.
10. Thread cutting.
11. Drilling.
12. Reaming.
13. Boring.
14. Tapping.

FACING: Facing is the operation of machining the ends of a piece of work to produce flat surface square with the axis. The operation involves feeding the tool perpendicular to the axis of rotation of the work.



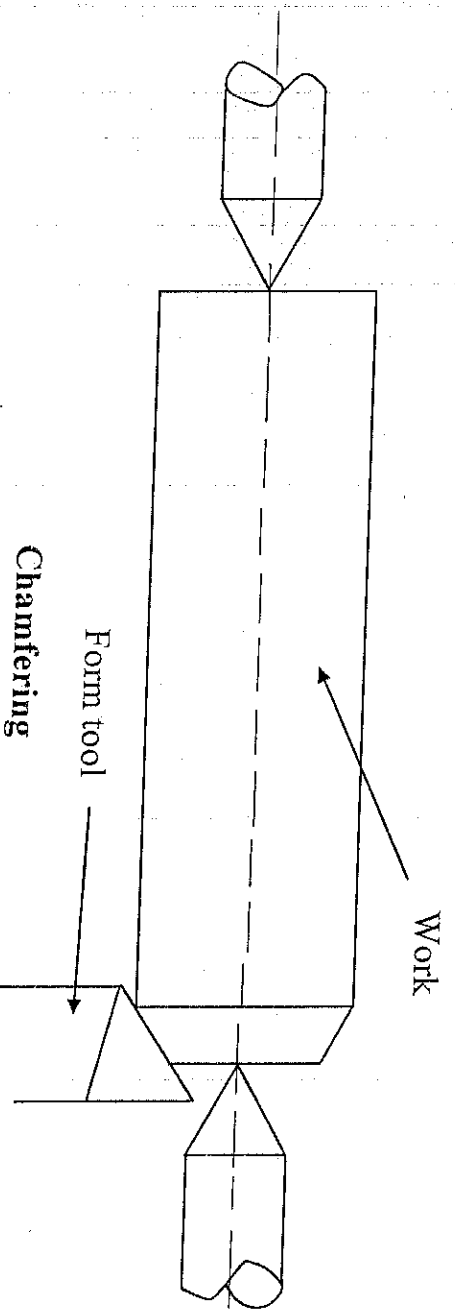
TURNING: Turning in a lathe is to remove excess material from the workpiece to produce a cylindrical surface of required shape and size.

STRAIGHT TURNING: The work is turned straight when it is made to rotate about the lathe axis and the tool is fed parallel to the lathe axis. The straight turning produces a cylindrical surface by removing excess metal from the workpiece.

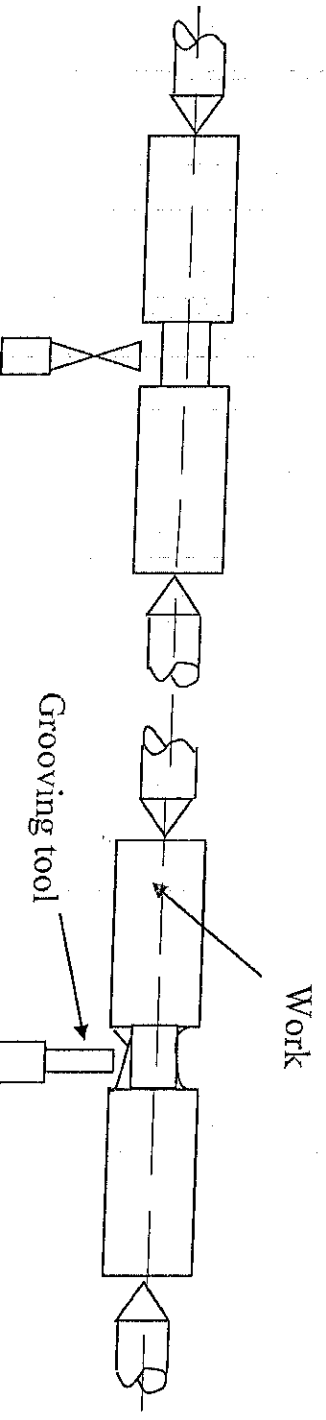


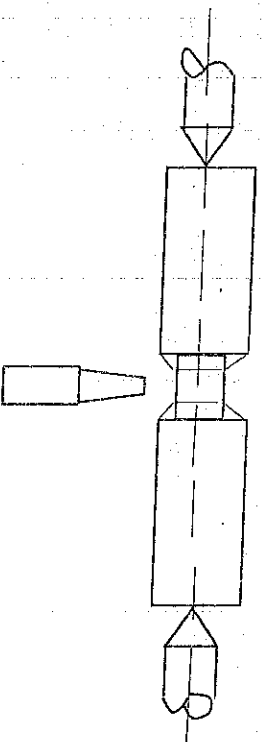
STEP TURNING: Step turning is the process of turning different surfaces having different diameters. The work is held between centres and the tool is moved parallel to the axis of the lathe. It is also called shoulder turning.

CHAMFERING: Chamfering is the operation of beveling the extreme end of the workpiece. The form tool used for taper turning may be used for this purpose. Chamfering is an essential operation after thread cutting so that the nut may pass freely on the threaded workpiece.



GROOVING: Grooving is the process of cutting a narrow grooving on the cylindrical surface of the workpiece. It is often done at the end of a thread or adjacent to a shoulder to leave a small margin. The groove may be square, radial or beveled in shape.





Grooving

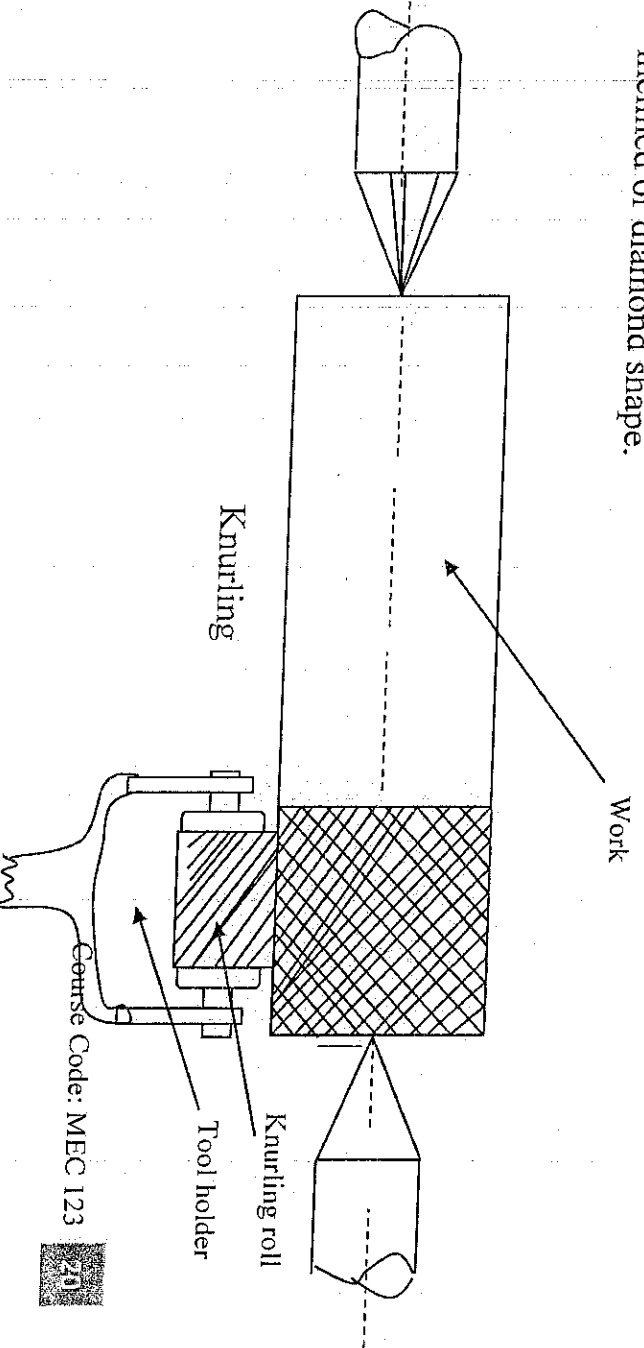
FORMING

Forming is a process of turning a convex, concave or any irregular shape. For turning a small length formed surface, a forming tool having cutting edges conforming to the shape required is fed straight into the work.

KNURLING

Knurling is the process of embossing a diamond shaped pattern on the surface of the workpiece. The knurling tool holder has one or two hardened steel rollers with edges of required pattern. The tool holder is pressed against the rotating work. The rollers emboss the required pattern. The tool holder is fed automatically to the required length.

Knurls are available in coarse, medium and fine pitches. The pattern may be straight, inclined or diamond shape.



The purpose of Knurling is:

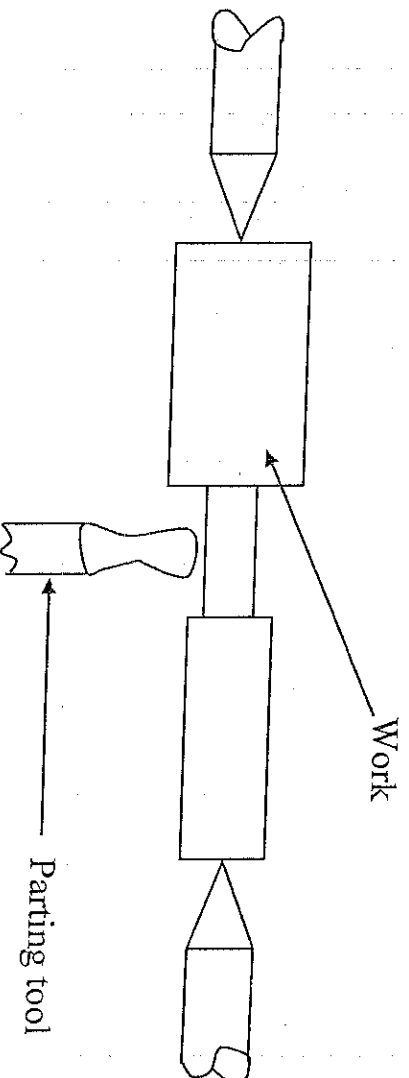
1. To provide an effective gripping surface.
2. To provide better appearance to the work.
3. To slightly increases the diameter of the work.

UNDERCUTTING

Undercutting is done:

- At the end of a hole.
- Near the shoulder of stepped cylindrical surfaces.
- At the end of the threaded portion in bolts.

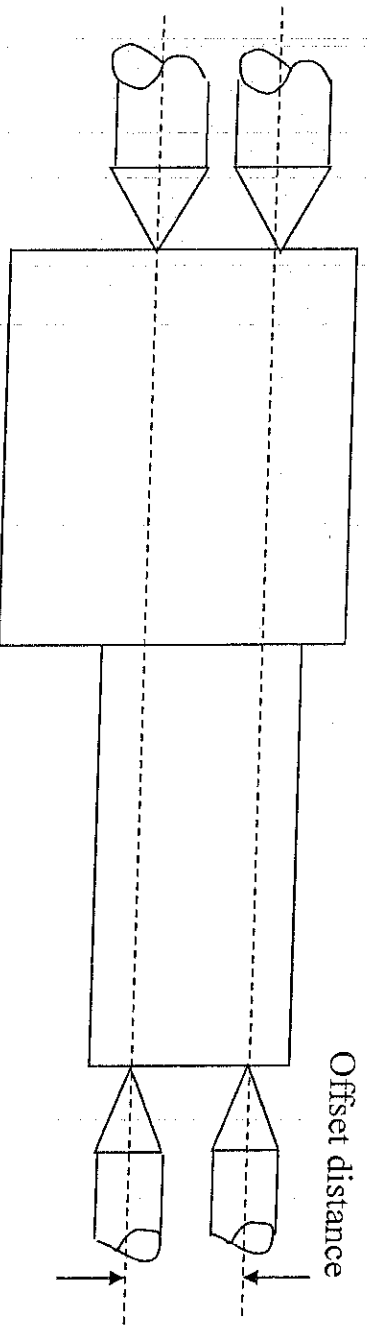
It is a process of enlarging the diameter if done internally and reducing the diameter if done externally over a short length. It is useful mainly to make fits perfect. Boring tools and parting tools are used for this operation.



Undercutting

ECCENTRIC TURNING

If a cylindrical workpiece has two separate axes of rotating, one being out of center to the other, the workpiece is termed as eccentric and turning of different surfaces of the workpiece is known as eccentric turning. The distance between the axis is known as offset. Eccentric turning may also be done on same special machines. If the offset distance, is more, the work is held by means of special machines. If the offset between the centers is small, two sets of centers are marked on the faces of the work. The work is held and rotated between each set of centers to machine the eccentric surface.



Eccentric turning

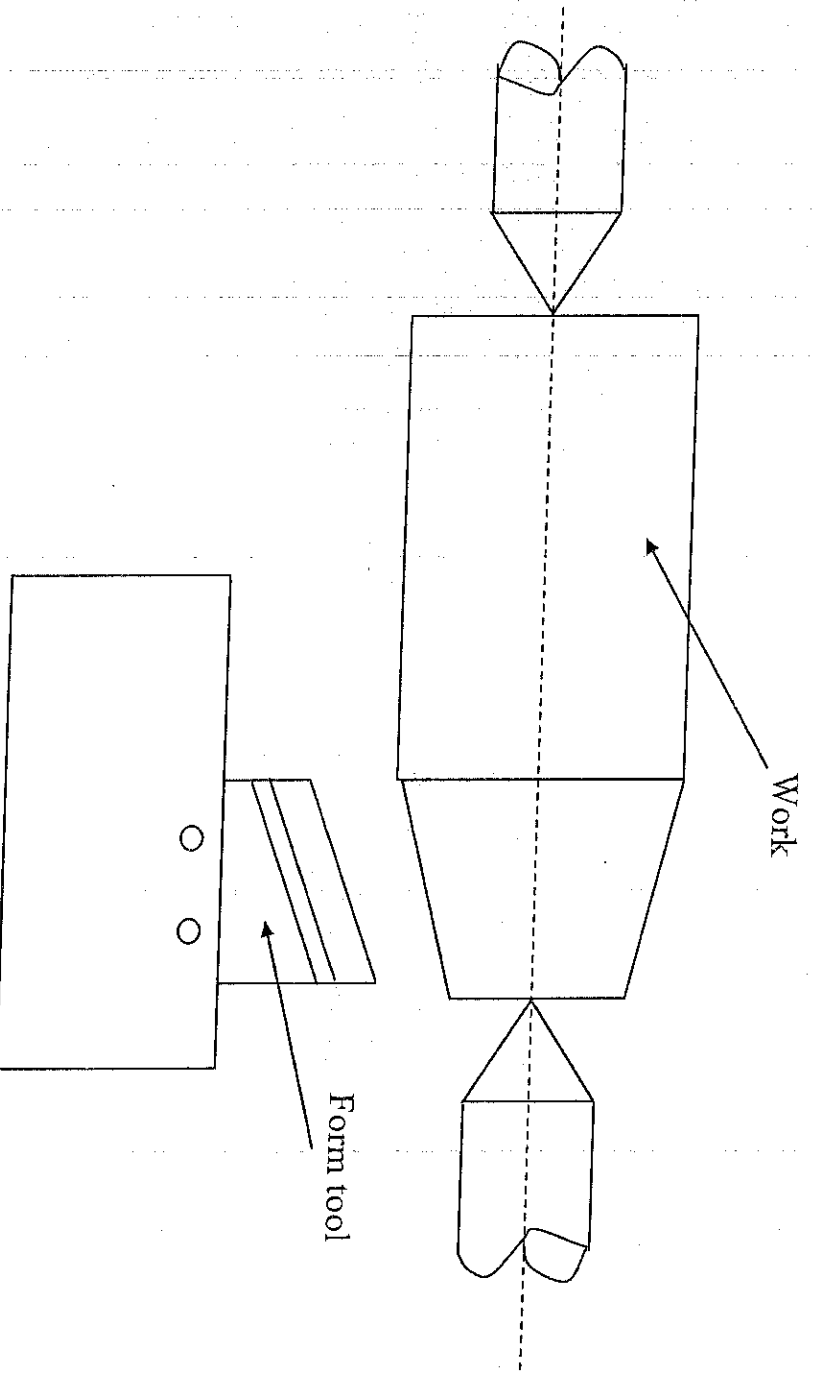
TAPER TURNING

A taper may be defined as a uniform increase or decrease in diameter of a piece of work measured along its length.

Taper turning methods:

1. Form tool method.
2. Compound rest method.
3. Tailstock setover method.
4. Taper turning attachment method.
5. Combine feed method.

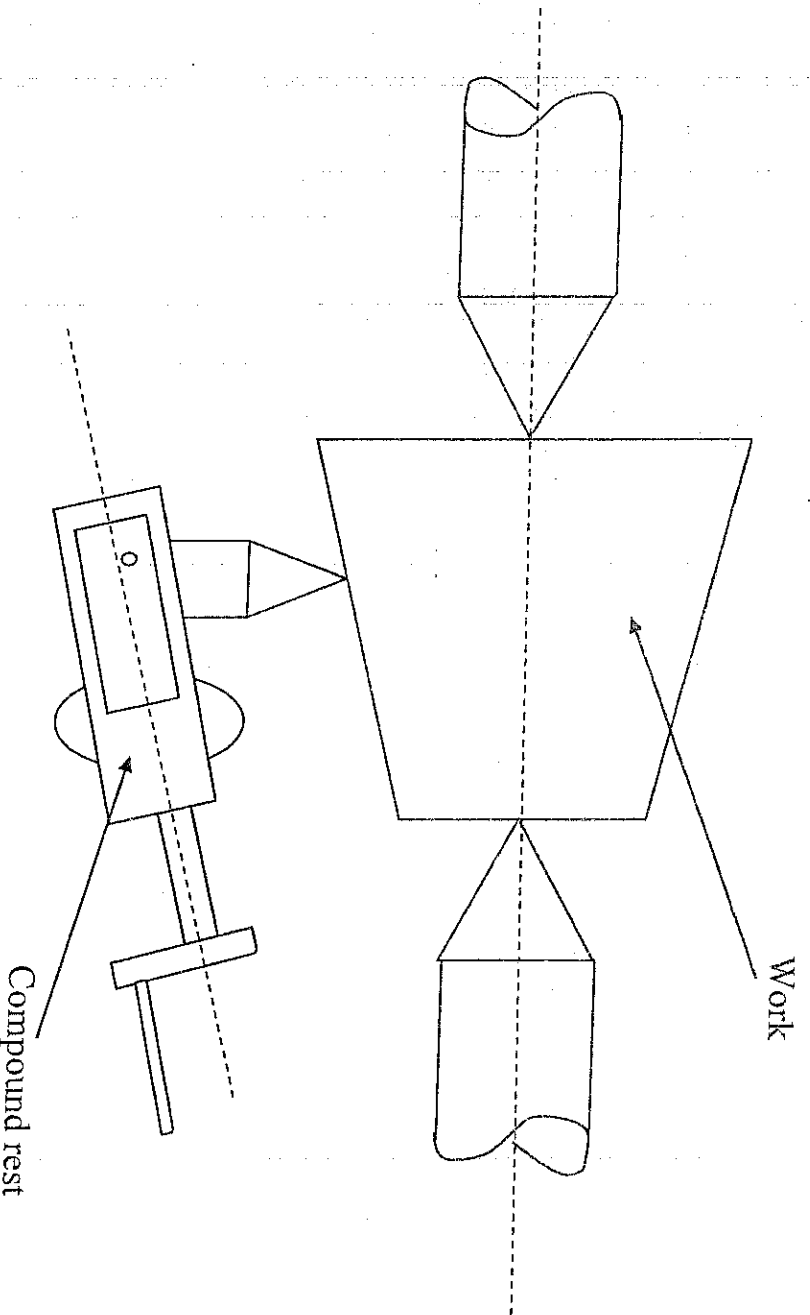
Form Tool Method: A broad nose tool is ground to the required length and angle. It is set on the work by providing feed to the cross-slide. When the tool is fed into the work at right angles to the lathe axis, a tapered surface is generated. This method is limited to turn short lengths of taper only. The length of the taper is shorter than the length of the cutting edge. Less feed is given as the entire cutting edge will be in contact with the work.



Taper turning by form tool method

Compound Rest Method: The compound rest of the lathe is attached to a circular base graduated in degrees, which may be swiveled and clamped at any desired angle. The angle of taper is calculated using the formula.

The compound rest is swiveled to the angle calculated as above and clamped. Feed is given to the compound slide to generate the required taper.



Paper Turning by Compound Rest Method

Tailstock Setover Method: Turning taper by shifting the axis of rotation of the workpiece at an angle to the lathe axis and feeding tool parallel to the lathe axis. The construction of tailstock is designed to have two parts namely the base and the body. The base is fitted on the bed guideways and the body having the dead centre can be moved at cross to shift the lathe axis.

The amount of setover – S, can be calculated as follows:

$$S = L \times \frac{D-d}{2l}$$

Where: D – larger diameter

L – Length of the work

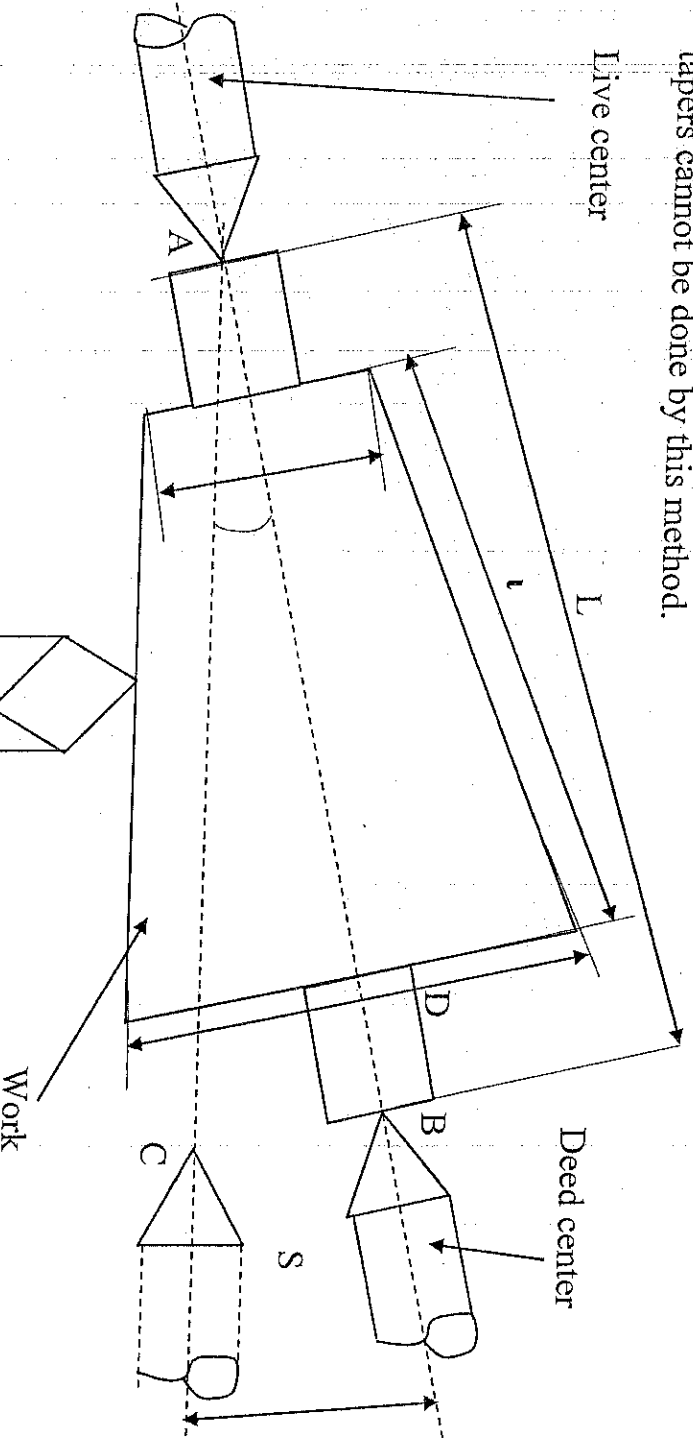
S – Amount of setover

d – Small diameter

l – Length of the taper

The dead center is suitably shifted from its original position to the calculated distance. The work is held between centers and longitudinal feed is given by the carriage to generate the taper.

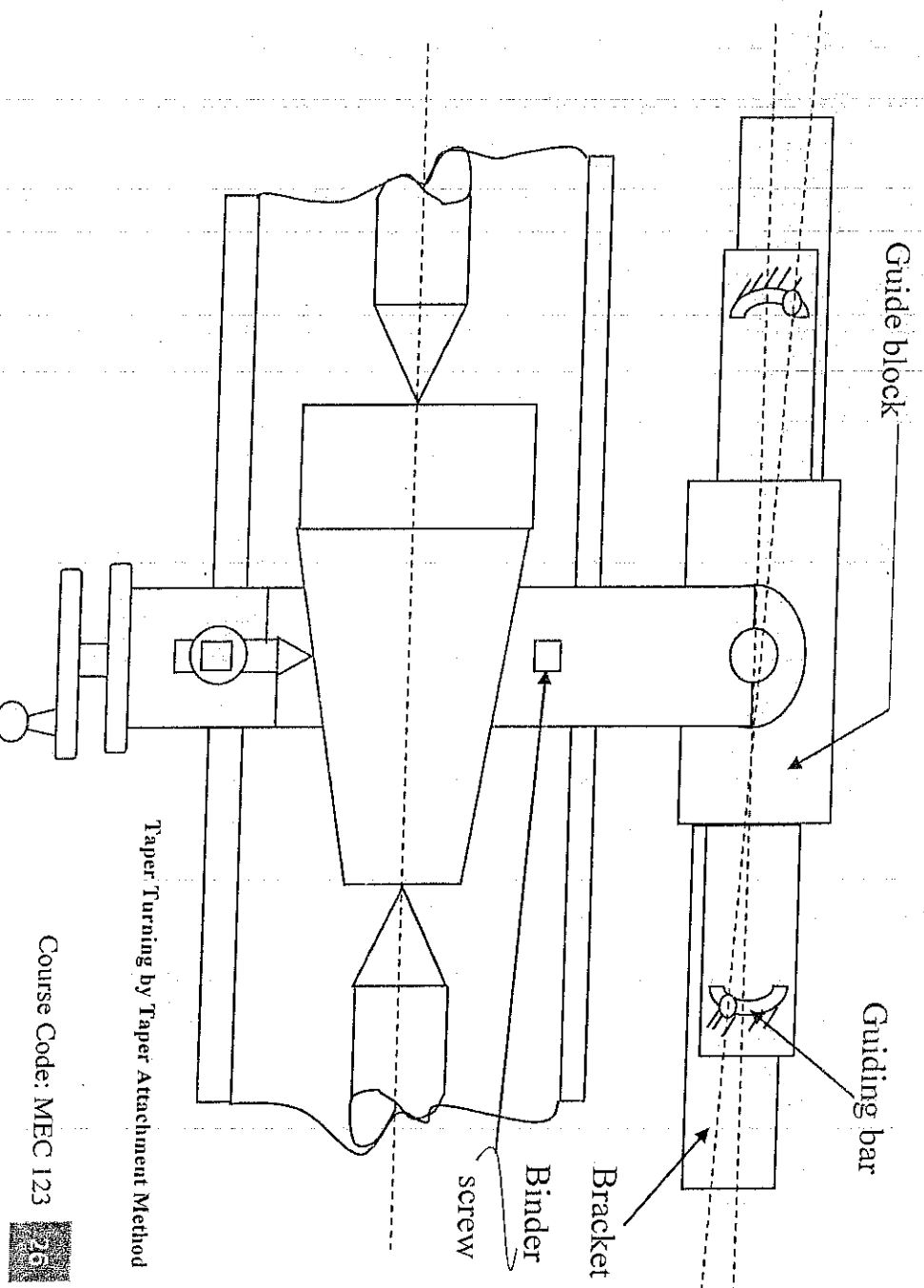
The advantage of this method is that the taper can be turned to the entire length of the work. Taper threads can also cut by this method. The amount of setover being limited, this method is suitable for turning small tapers (approx. up to 8°). internal tapers cannot be done by this method.



Taper Turning by Tailstock Setover Method

Taper Attachment Method: The taper attachment consists of a bracket which is attached to the rear end of the lathe bed. It supports a guide bar pivoted at the center. The bar having graduation in degrees may be swiveled on either sides of the zero graduation and set at the desired angle to the lathe axis. A guide block is mounted on the guide bar and slides on it. The cross slide is made free from its screw by removing the binder screw. The rear end of the cross slide is tightened with the guide block by means of a bolt. When the longitudinal feed is engaged, the tool mounted on the cross slide will follow the angular path as the guide block will slide on the guide bar set at an angle of the lathe axis. The depth of cut is provided by the compound slide which is set parallel to the cross-slide.

The advantage of this method is that long tapers can be machined. As power feed can be employed, the work is completed at a shorter time. The disadvantage of this method is that internal tapers cannot be machined.



Combined Feed Method: Feed is given to the tool by the carriage and the cross-slide at the same time to move the tool at resultant direction to turn tapers.

✶ **Thread Cutting:** Thread cutting is one of the most important operations performed in a lathe. The process of thread cutting is to produce a helical groove on a cylindrical surface by feeding the tool longitudinally.

1. The job is revolved between centres or by the longitudinal feed should be equal to the pitch of the thread to be cut per revolution of the workpiece.
2. The carriage should be moved longitudinally obtaining feed through the leadscrew of the
3. A definite ratio between the longitudinal feed and rotation of the headstock spindle should be found suitable gears with required number of teeth should be mounted on the spindle and the leadscrew.
4. A proper thread cutting tool is selected according to the shape of it is mounted on the tool post with its cutting edge at the lathe axis and perpendicular to the axis of the work.
5. The position of the tumbler gears are adjusted according to the type of the thread (right hand or left hand).
6. Suitable spindle speed is selected and it is obtained through back.
7. Half nut lever is engaged at the right point as indicated by the thread chasing.
8. Depth of cut is set suitably to allow the tool to make a light cut on the workpiece.
9. When the cut is made for the required length, the half nut lever is the carriage is brought back to its original position and the above procedure is repeated until the required depth of the thread is achieved.
10. After the process of thread cutting is over, the thread is checked by suitable gauges.

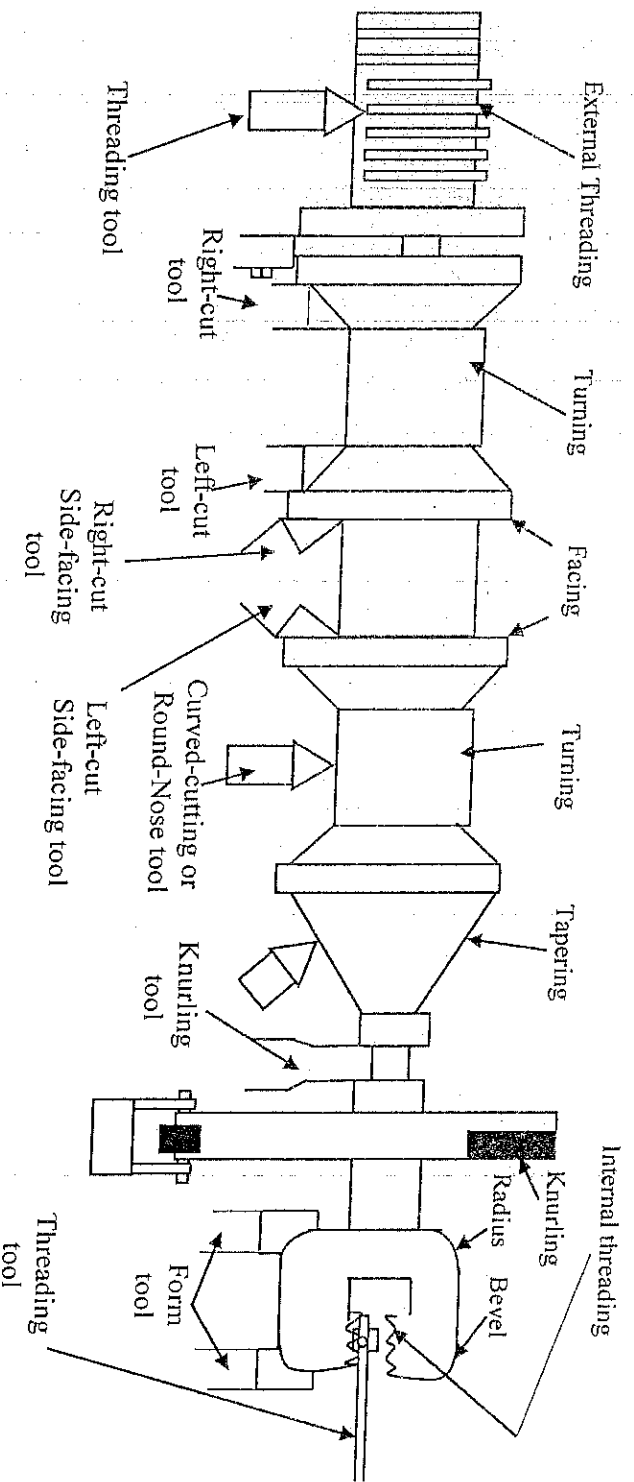
THE LATHE (TURNING MACHINE)

A lathe machine is the most versatile of all the machine tools in use in the production of parts and components in all production industries and ~~low~~^{shop}-rooms/machine-shops. The lathe is the first runner of all machine tools, the first application of the lathe principle was probably the potter's wheel. This machine rotated a mass of clay and enabled the ~~day~~^{clay} to be formed into a cylindrical shape, the modern lathe operates on the same basic principles.

TYPES OF LATHE

1. **Centre Lathe:** Is the general purpose lathe, it can vary in size and complexity but the most useful kind is the sliding, surfacing and screw cutting lathe which is the most generally used in school for training.
2. **Turret Lathes/ Automatic:** Is a semi-automatic lathe to make parts in greater quantities, respective work is often confined into them for mass production.
3. **Heavy Duty Lathe:** These are known as larger-sized lathe and is used for roughing and finishing cut, and also large component are turned on it such as engine blocks and a very long shaft.
4. **Instrument Lathes:** These lathes come in the category of all purpose, low-production lathe and are used by instrument maker.
5. **Facing Lathes:** These are used to machine workpiece of large diameter, but short in length in a single piece production and repair-ship such as wheel, gear, pulley, etc.
6. **Speed Lathe:** Is a lathe in its simplest form, the speed of these lathes are much higher as compared to conventional lathe such lathes are used for making center hole in a workpiece prior to holding them between centers for metal spinning and wood work.

COMMON LATHE OPERATION



Common Lathe Operations

Common Lathe Operations

- 7. Knurling:** Is a process of impressing a diamond shaped straight line pattern into the surface of the workpiece to improve its appearance to provide a better gripping surface. Straight knurling is often used to increase the workpiece diameter when a press fit is required.

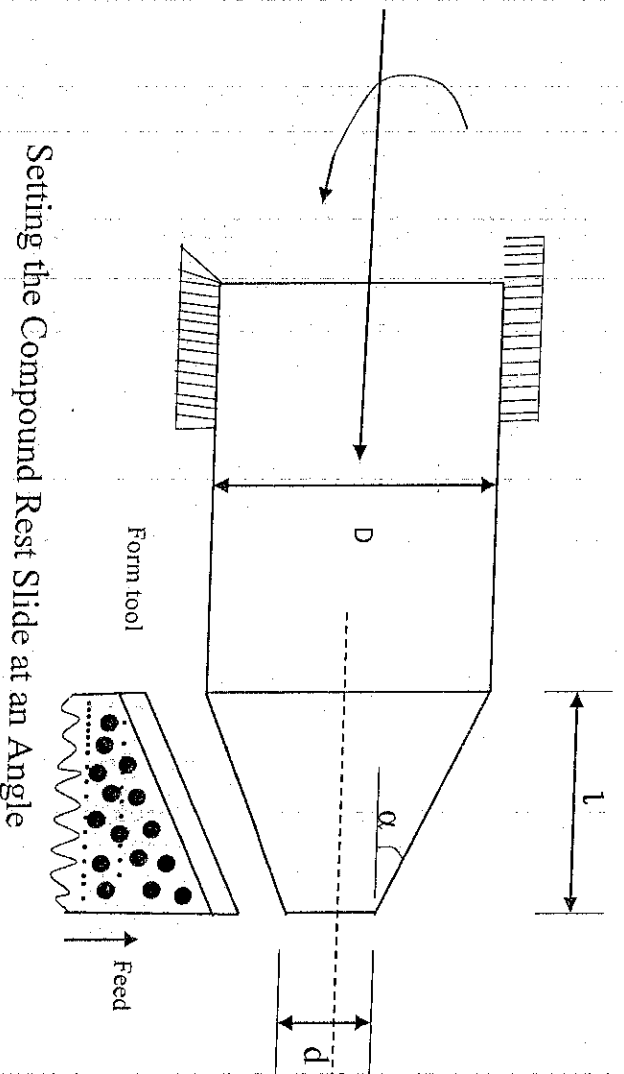
- 8. Thread Cutting:** Is one of the most interesting operations performed on lathe. It involves manipulation of the lathe part co-ordination of the hands and strict attention to the operation both internal and external thread can be cut on a lathe.

- 9. Taper Turning:** There are four (4) method of turning a taper on a workpiece:

- Using the tool angle.
- Setting compound rest slide at an angle.
- Setting over the tailstock.
- Using a taper turning attachment.

(i) Using The Tool Angle

The cutting edge of the tool is set round to the angle required and the cut is taken by feed straight, this method can be used only for short tapers.



(ii) Setting The Compound Rest Slide at An Angle

The compound rest slide and lock at the required angle and tool fed by hand automatic feed lever. This method is used for short tapers.

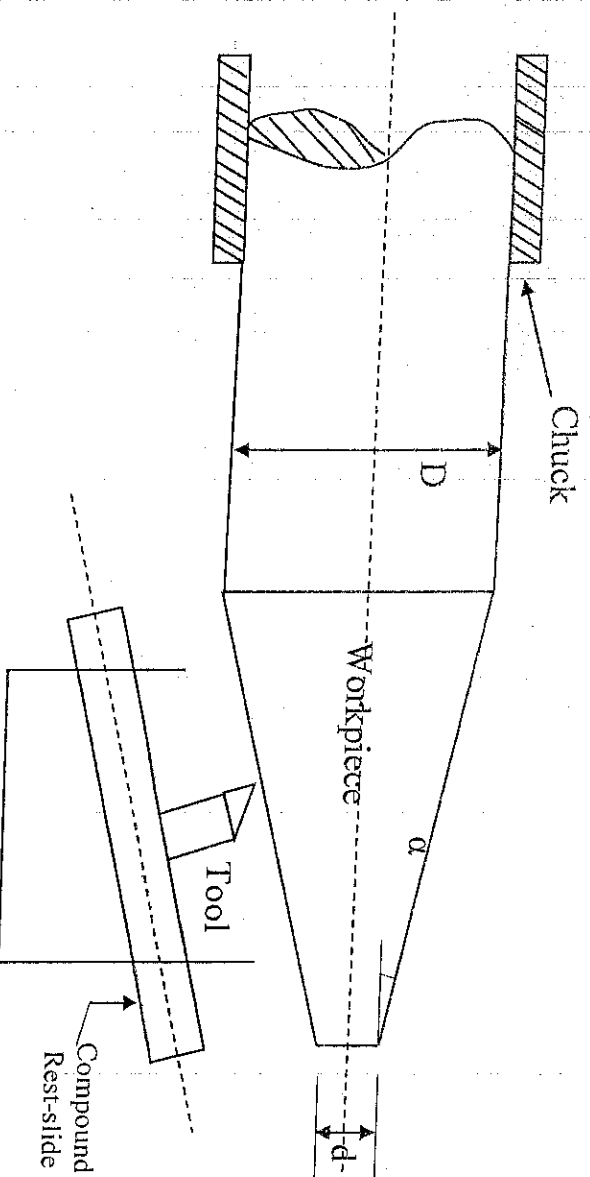
The angle is determined by $\tan \alpha = \frac{D - d}{2l}$

l = Length of taper.

D = Bigger diameter

d = Small diameter

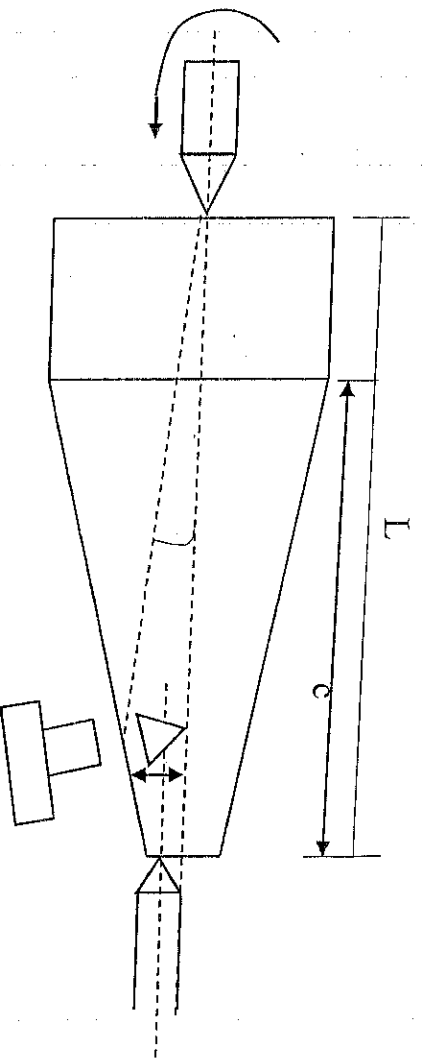
α = Half taper angle



(iii) Setting Over The Tailstock

This method is used only for a long taper, the tailstock is intentionally set out of line with the head stock then the work becomes angled to the direction of movement of the carriage and will be tapered in proportion to the amount of set over. It can be determined by

$$h = L \frac{(D - d)}{2l}$$



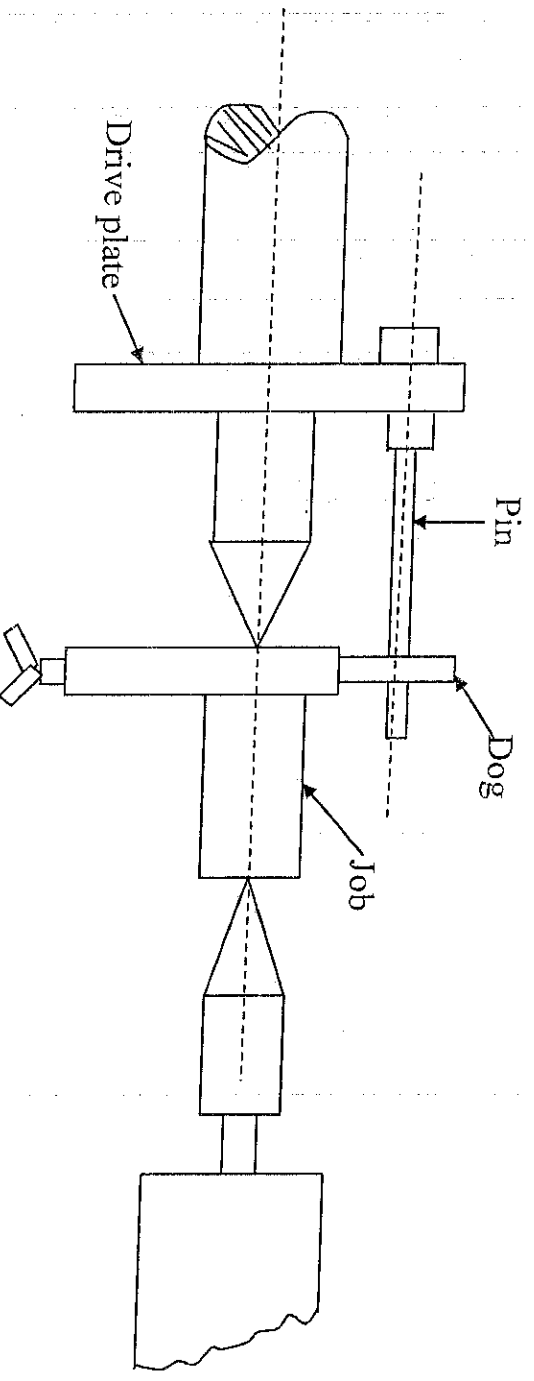
(iv) Using Taper-Turning Attachment

These can be used on the lathe constructed or specially adopted for the purpose, their use is the best way of turning long tapers because the work runs finely on the centers all the time. This attachment used to be mounted on a lathe bed, at its bar can be turned to the required angle and clamped in place.

WORK HOLDING DEVICES/ LATHE ATTACHMENT

The common work holding devices used for a centre lathe are:

- (i) **Centres:** Workpiece (shafts and axis) with L/D ratio > 4 , are mounted between the head stock and tail stock centres. The end center of such workpiece must have been previously drilled and also driving dug must have been clamped on one end.



Milling machines are machine tools used to produce one or more machined surfaces accurately on a workpiece ~~on a lathe~~ materials. i.e. the workpiece.

WORKPIECE AND CUTTER HOLDING DEVICE AND ATTACHMENT USED ON THE MILLING MACHINE

Milling Machine attachment may be divided into three (3) Classes:

1. Those designed to hold special attachments, they are attached vertical, universal, rack milling and slotting attachments, these attachments are designed to increase the versatility of the machine.
2. Those designed to hold standard cutters e.g. Arbors collets, adapters. Arbors are used for mounting milling cutters, collect and adapters are used for mounting drills or other tapered shank tools in the main spindle of the machine or the vertical milling machine.
3. Those designed to hold the workpiece such as vise rotatory table, indexing or dividing head vices. These are used most widely in work holding devices for milling and they are of different types. Indexing or dividing head: This is very useful accessing that permits the cutting of bolt head, gear teeth etc.

✶ TYPES OF MILLING CUTTERS AND THEIR USES.

1. **Plain Milling Cutters:** These cutters are cylindrical in shape and have cutting teeth on periphery only. They are used for milling plain or flat surface.
2. **Face Milling Cutters:** These have teeth on end face, and are used for machining large flat surface.
3. **Profile Milling Cutters:** Some common shape of these cutter are convex or concave. They are used to machine any standard gear teeth.
4. **End Mills Cutters:** Have cutting teeth on the end as well as on the periphery. They are employed for making deep grooves, steps, perpendicular planes, etc.
5. **Angular Milling Cutters:** These are made as single or double angle cutters and are used to machine angles other than 9.

6. **T-Slot:** This is used to cut the wide horizontal groove at the bottom of a T-Slot after the arrow vertical groove have been machined with an end will.

FEATURE AND WORKING PRINCIPLE OF DIVIDING HEAD

Dividing Head is a one of the most important attachment for milling machine. It is used to provide the circumference of workpiece into equal spaced division when milling gears, square, hexagon, etc.

METHODS OF INDEXING

The following are the various method of indexing:

- a. Direct.
- b. Simple.
- c. Differential.
- d. Angular.

MILLING OPERATION

Milling is a machining operation in wheels each tool of a multi-toothed rotating cutter engages, the workpiece in turn and removes a relatively a small amount of metal in the form of chips. Milling operation can be widely classified as end-cutting devices, a milling machine is normally used in producing plane surfaces, flat and square to each other by suitable combination of cutter workpiece movements it could be manually or mechanically operated to achieve the desired result. Angular and semi-circular surfaces can as well be achieved on a milling machine.

FEED AND CUTTING SPEED

Feed Speed: Is the rate at which the tool is moved into the part or the part into the tool, or it is the rate at which the work moves in the revolving cutter. Feed is measured in feet/ inches or millimeters per time period.

Feed speed = No of cutter teeth x Feed/tooth x cutter/ mins. C_{pT} or F_{pT} is defined as the amount of material that should be removed by each tooth of the cutter as it revolves and advances into the work.

$$\text{Feed speed} = N \times C_{pT} \times 4C_s = \frac{4C_s N(C_{pT})}{D}$$

Cutting Speed (also called surface speed or simplify speed): May be defined as the rate at the workpiece surface, irrespective of the machining operation used. Or the relative speed between the cutting tool and workpiece is known as the cutting speed and is expressed in meters per minute. In milling and drilling, it is the surface speed of the cutting tool.

$$\text{Cutting speed } S = \frac{\pi d N}{1000} \text{ m/min.}$$

$$\text{or } S = v_R = \pi D N \text{ (mm/see)}$$

$$\text{or } C_s = \frac{ND}{4} \text{ m/min}$$

FACTORS WHICH INFLUENCE CUTTING SPEED DETERMINATION FOR MILLING:

1. The type of cutting tool material being used.
2. The type of material being machined.
3. The depth of cut.
4. Feed rate.
5. Type of coolant to be used.
6. Type of finish to be obtained
7. The diameter of cutter.
8. The rigidity of machine and work setup.

The table below are the recommended feed per tooth (High Speed Cutter)

Materials	Face mill (mm)	Helical mill (mm)	Slotting & Side mill (mm)	Face mill (inches)	Helical mill(inches)	Slotting mill(inches)
Alloy steel	0.15	0.12	0.1	0.006	0.005	0.004
Aluminum	0.55	0.45	0.33	0.022	0.18	0.13
Cast Iron	0.33	0.25	0.18	0.013	0.010	0.007
Brass & Bronze	0.35	0.28	0.2	0.014	0.011	0.008
Machine Steel	0.3	0.25	0.18	0.012	0.010	0.007

SAFETY PRECAUTION TO BE OBSERVED WHEN MILLING

1. Be sure that the work and the cutter are mounted securely.
2. Always wear safety glasses.
3. When mounting or removing mill cutters, always hold them with a cloth to avoid being cut.
4. When setting up work, move the table as far as possible from the cutter to avoid cutting your hands.
5. Never attempt to mount, measure or adjust work until the cutter has completely stopped.
6. Always use a brush, not rag to remove cuttings after the cutter has stopped revolving.

Example 1: Find the cutting speed of 50mm diameter bar being turned with a spindle speed of 178rev/min.

Solution

$$\begin{aligned} \text{From cutting speed} &= \frac{\pi dN}{1000} = \frac{\pi \times 50 \times 178}{1000} \\ &= 28\text{m/min} \end{aligned}$$

Example 2: Find the feed speed in inches per minute using a 3.5 in diameter, 12-tooth helical cutter to cut machine steel (CS 80).

Solution

$$\text{From } N(\text{in/min}) = \frac{N \times \text{CPT} \times 4\text{Cs}}{D} = \frac{4 \times 80}{3.5} = 91 \text{ rev/min}$$

$$\begin{aligned} \text{Feed speed (in/min)} &= N \times \text{CPT} \times \text{rev/min.} \\ &= 12 \times 0.010 \times 91 \\ &= 10.9 \approx 11 \text{ in/min.} \end{aligned}$$

Example 3: At what speed should a 2-in diameter carbide cutter revolve to mill a piece of cast iron (Cs 150).

Solution

Given data

Where $D = 2 \text{ in}$

$\text{Cs} = 150$

$$\begin{aligned} \text{From } N(\text{rev/min}) &= \frac{4\text{Cs}}{D}, \\ &= \frac{4 \times 150}{2} \\ &= 300 \text{ rev/min} \end{aligned}$$

But in mm/min, the formula for calculating cutter speed is rev/min

$$\begin{aligned} &= \frac{320\text{Cs}}{\pi} \\ &= \frac{320 \times 150}{\pi} \\ &= 15278.87 \text{ mm/min} \end{aligned}$$

Assignment

1. Calculate the rev/min required for a 75mm diameter high speed steel milling cutter when cutting machine steel (Cs 30m/min).
2. A turning operation has to be performed on an aluminum rod of diameter 50mm and length 300mm. The spindle speed of lathe is given to be 500RPM. The feed and depth of cut are 0.15mm/rev and 0.3mm respectively. Draw a neat sketch of the turning operation described above. Find out the cutting speed in mm/s and volumetric material removal rate (MRR_v).

GRINDING

Grinding is an important part of the machine tool trade: improved grinding machine construction has permitted the production of parts to extremely fine tolerances with improved surface finishes and accuracy.

TYPES OF GRINDING MACHINE (m/c)

There are three principal types of grinding machine namely:

- i. Surface grinding machine
- ii. Cylindrical grinder
- iii. Universal cutters and tool grinder

Surface Grinding Machine: Is used primarily for grinding flat surfaces on hardened or unhardened workpiece, since workpiece can be held by various methods and the wheel-face can be shaped by dressing. It can perform various operation like form, angular and vertical grinding, surface grinding brings the work to close tolerances and produce a high surface finish.

Cylindrical Grinders: Is used to grind the diameter of workpiece accurately to size and to high surface finish. It can also be used to produce cylindrical, tapered and multidiameter workpiece.

Universal Cutter and Tool Grinder: Is designed primarily for grinding operations to be performed. Other operation that may be performed are cylindrical, taper surface grinding etc.

MAIN FEATURES OF GRINDING MACHINE

1. It can improve surface finish and accuracy of workpiece.
2. It eliminates the need for conventional machine.
3. It eliminates the need for other machining process.
4. It is extensively used in production of both hardened and unhardened parts where accuracy and surface finish is required.
5. Modern grinding machines permit the manufacture of intricate parts faster and more accurately than other machining methods do.

STRUCTURES OF GRINDING WHEEL

The structure of grinding wheel is the space relationship of the grain and bonding material to the voids that separate them, in brief, it is the density of the wheel; if the spacing of the grains is close, the structure is dense. If the spacing of the grains is relatively wide, the structure is open.

Selection of the wheel structure depends on the type of work required, wheel with open structure provides greater chip clearance than those with dense structures and remove material faster than dense wheel.

Diagram

Assignment 2

Explain fully with the aid of diagram where necessary how the following operations can be performed on grinding machine:

- (1) Surface grinding machine.
- (2) Taper grinding operation.
- (3) Tool and cutter grinding operation.
- (4) Centerless operation.
- (5) Cutting-off operation.

WHEEL TESTING AND MOUNTING

The wheel must be tested to see that they have not been damaged in transit. They should be suspended and tapped lightly with a screw driver handle for a small-wheels or with a wooden mallet for large wheels.

If vitrified or silicate wheel are found they give clear metallic ring, organic-bonded wheel gives a duller ring and checked wheels do not produce a ring. Wheel must be dry and free from saw dust before testing otherwise the sound will be deadened.

One way of mounting of a grind wheel is shown below, it is clamped by flanges (one each on its either side) directly on the spindle. Thick compressible washers (0.5 to 3mm thick) made of card-board, rubber or leather etc. should be inserted in between the flanges and the wheel.

Diagram

WHEEL BALANCING OR BALANCING OF GRINDING WHEEL

Proper balance of a mounted grinding wheel is very important, since improper balance will greatly affect the surface finish and accuracy of the work. There are two methods of balancing wheel:

1. **Static Balancing:** On some grinders, the wheel is balanced off the machine with the use of a balancing stand and arbor. The wheel is mounted at the centre of perfectly straight and round arbor, the assembly is then rested on straight knife edged ways or roller rests of the balance machine as shown below, out of balanced wheel will not remain stationary. It will come to rest with its heavy side underneath. The wheel is balanced by cutting some of the lead from the heavy side of the bush.
2. **Dynamic Balancing:** Most new grinding machine are equipped with ball bearing balancing devices, which automatically balance a wheel in a matter of seconds while it is revolving on the grinder.

TRUING AND DRESSING A GRINDING WHEEL

Truing: Is the process of making a grinding wheel round and concentric with its spindle axis and produced the required shape on the wheel. This procedure involves the grinding or wearing away of a portion of the abrasive section of the grinding wheel in order to produce desired shape.

Dressing: Is the operation of removing the dull grains and metal particles. This operation exposes sharp cutting edge of the abrasive grains to make the wheel cut better.

A dull glazed or loaded wheel should be dressed for the following reasons:

- i. To reduce the heat generated between the work surfaces.

4.6 CONSTRUCTION OF THE BOX CULVERT

CTHSE1

MR PHILLIP MATHSEN

The construction of the box culvert involved preliminary survey, provision for a solid foundation, assembling the precast concrete elements in-situ and casting the additional parts required for the completion of the element. All safety rules and regulations were adhered to and PPEs were worn by everybody involved in the construction including the trainee.

Processes involved in the construction include:

1. Topography survey,
2. Site clearance,
3. Earthworks,
4. Foundation (stone base and concrete blinding),
5. Fabrication of precast concrete walls and slabs,
6. Installation of precast elements (walls),
7. Casting of in-situ bottom slabs,
8. Casting of in-situ portion of top slabs,
9. Priming of surface,
10. Drop structure,
11. French drain,
12. Backfilling and compaction.

TEST 2

Q1. List two (2) methods of wheel balancing.

Q2. Mention four (4) methods of indexing.

Q3. Calculate the indexing and change gears required for 57 divisions. The change gears supplied with the dividing head are as follows: 24, 28, 32, 40, 44, 48, 56, 64, 72, 86, 100. The available index plate hole circles are as follows:

plate 1: 15, 16, 17, 18, 19, 20

plate 2: 21, 23, 24, 29, 31, 33

plate 3: 37, 39, 41, 43, 47, 49

Q4. Mention any five (5) types of cutting tool and draw a neat diagram for each.

Sol

$$\text{Indexing} = \frac{40}{N} = \frac{40}{57}$$

Since, there is no 57-hole circle and it is impossible to reduce this fraction to suit any hole circle, it is necessary to select an approximate number close to 57, for which simple indexing may be calculate.

— Let the approximate number of divisions equal 56

— Indexing for 56 divisions = $\frac{40}{56} = \frac{5}{7}$ or 18 holes in the 21-hole circle

$$\text{Gear ratio} = (A - N) \frac{40}{A} = (56 - 57) \times \frac{40}{56} = -\frac{40}{56} = -\frac{5}{7}$$

$$\text{Change gears} = -\frac{5}{7} \times \frac{8}{8} = -\frac{40(\text{Spindle})}{56(\text{worm gear})}$$

Therefore, for indexing 57 divisions, a 40-tooth gear is mounted on the dividing head spindle, and a 56-tooth gear is mounted on the worm shaft. Since the fraction is a negative quantity and simple gearing is to be used, the index plate rotation is negative, or counterclockwise, and two idler must be used. After the proper gears are installed, the simple indexing procedure for 56 divisions should be followed.

Class tutorials

Structure of grinding wheel

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